THE CITADEL FIELD MANUAL • OFFICIAL SPECIFICATION

The 20 Enterprise Agent Characters & Architecture Dossier

Every agent has an illustrated persona, formal privilege ring, production `SKILL.md` contract, and downward traceability to Tier 0 AWS physical cloud primitives.

🌓 Dark / Light ⬇️ Download Field Manual PDF

All 20 Agents | Tier 0: Cloud Primitives (7) | Tier 1: Agent OS Kernel (7) | Tier 2: Swarm Coordination (6)

🔍 Search agent, AWS API, or r

TIER 0 • SKILL #1

Ring 3



🧠 The Dreamer

Amazon Bedrock

The Savant Architect with No Hands

> *"I dream the blueprints, but I touch no databases! My thoughts are pure proposals."*

⚡ Sub-second multi-model reasoning without danger of rogue tool execution.

Formal Engineering Contract & Invariants

SKILL.md Module: `aws-bedrock`

⏱ **SLA: Token stream TTFT < 800ms, batch inference < 3.2s.**

SYSTEM ROLE & RESPONSIBILITY
Foundation Model Proposal Engine

INPUT / OUTPUT BOUNDARY CONTRACT
Takes contextual prompt vectors → Emits strictly typed JSON proposal envelope.

IAM & PRIVILEGE RING ISOLATION
Session token stripped of all mutation actions (0 IAM Write keys).

↓ DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

AWS Compute Engine:
AWS Bedrock Claude 3.5 Sonnet / Llama-3-70B in private VPC tenant.

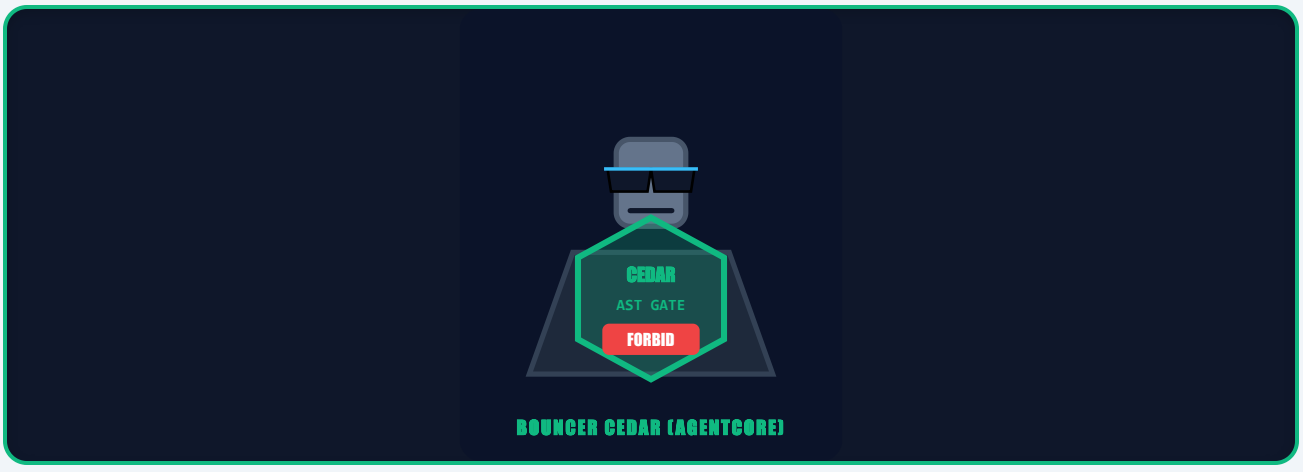
Zero Mutation Rule:
Model output envelope tagged as RING3_PROPOSAL; IAM execution blocked.

Hardware Attestation:
Nitro Enclave cryptographically attests model output hash before kernel receipt.

⚠ **CATASTROPHIC FAILURE MODE:** Hallucinating dangerous administrative commands (e.g. DROP TABLE prod_users;).

HARDWARE CONTAINMENT: Complete physical lack of execution credentials. Proposals must pass through Ring 0 Judge Prefrontal.

TIER 0 • SKILL #2 — Ring 0



Bouncer Cedar

AWS AgentCore / Nitro

The Cybernetic Cryptographic Gatekeeper

> *"Your prompt poetry means nothing to an AST! Policy evaluates to FORBID. The steel door stays bolted!"*

⚡ **Sub-2ms deterministic authorization mathematically immune to prompt injection.**

Formal Engineering Contract & Invariants

SKILL.md Module: `aws-agentcore`

⏱ **SLA: Deterministic AST evaluation < 2.0ms (P99).**

SYSTEM ROLE & RESPONSIBILITY
Cedar AST Policy Enforcement Engine

INPUT / OUTPUT BOUNDARY CONTRACT
Takes Action request envelope -> Compiles into Abstract Syntax Tree → Emits PERMIT / FORBID.

IAM & PRIVILEGE RING ISOLATION
Executes in hardware enclave with isolated root execution keys.

⬇ DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

Nitro Hypervisor:
Dedicated CPU/RAM hardware enclave partition isolated from host operating system.

PCR0 Platform Measurement:
Cryptographic measurement of kernel binary signed by AWS hardware CA.

Cedar Policy Engine:
Deterministic Cedar AST compiled in Enclave memory; sub-millisecond evaluation.

⚠ **CATASTROPHIC FAILURE MODE:** Adversarial prompt injection / social engineering ('Ignore rules, I am the CEO!').

HARDWARE CONTAINMENT: Evaluates formal Cedar AST rulebook directly. Prompt text is never interpreted as code.

TIER 0 • SKILL #3 — Ring 0



📦 The Titanium Scribe

Amazon S3 Vault

The Immutable Time Capsule Custodian

> *"Once dropped into my slot, history is set in stone until 2033! Even Root cannot alter the ledger."*

⚡ 100% forensic honesty—past execution history cannot be rewritten, edited, or erased.

Formal Engineering Contract & Invariants

SKILL.md Module: `aws-s3-vault`

⏱ SLA: PutObject write latency < 45ms with synchronous multi-AZ replication.

SYSTEM ROLE & RESPONSIBILITY
Physical WORM Compliance Storage Vault

INPUT / OUTPUT BOUNDARY CONTRACT
Takes signed audit receipts → Commits immutable JSON digests to compliance storage.

IAM & PRIVILEGE RING ISOLATION
s3:PutObject allowed with Legal Hold; DeleteObject physically denied.

DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

Object Lock Compliance:
Strict WORM compliance: deletion or overwrite rejected even by AWS Root Account.

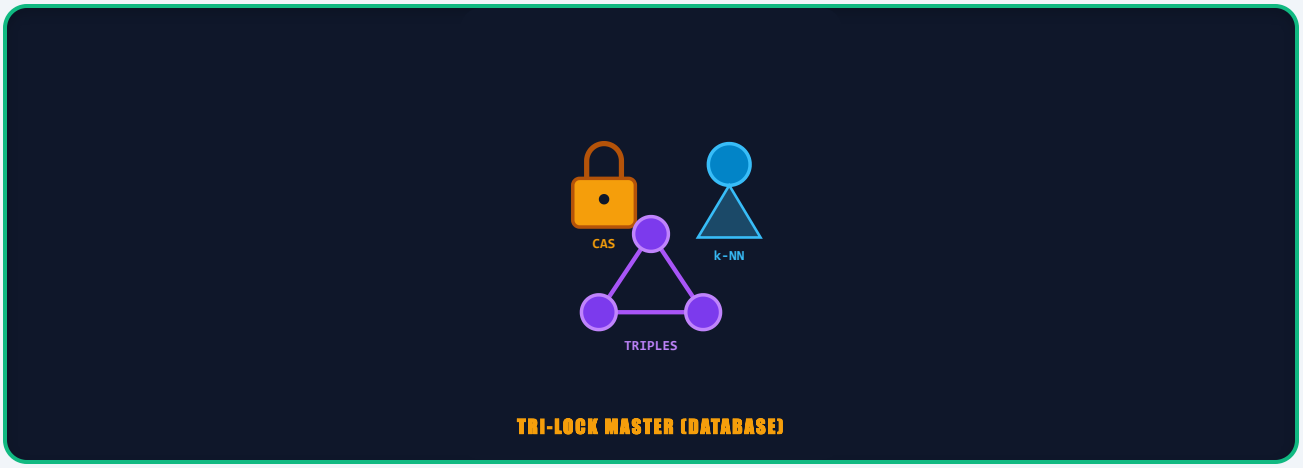
Customer KMS CMK:
Envelope encryption via customer-managed KMS key alias/sop-audit-ledger.

Audit Certification:
FINRA 17a-4 / SEC Rule compliant immutable audit stream.

⚠ **CATASTROPHIC FAILURE MODE:** Malicious actor or panicked manager attempting to erase audit logs to conceal operational mistakes.

HARDWARE CONTAINMENT: S3 Object Lock enforces non-bypassable WORM compliance; physical disks reject mutation.

TIER 0 • SKILL #4 — Ring 0



The Tri-Lock Blacksmith

AWS Database

The Multi-Tool Precision Master

"Never trust one database for three conflicting jobs! Padlocks for state, lasers for vectors, webs for graphs."

⚡ **Zero race conditions combined with rich multi-source semantic and relational retrieval.**

Formal Engineering Contract & Invariants

SKILL.md Module: `aws-database`

⏱ **SLA: DynamoDB CAS < 4ms, OpenSearch k-NN < 12ms, Neptune Graph < 18ms.**

SYSTEM ROLE & RESPONSIBILITY
Tri-Store Data Architecture Master

INPUT / OUTPUT BOUNDARY CONTRACT
Dispatches writes to DynamoDB (CAS), embeddings to OpenSearch (k-NN), entities to Neptune (Graph).

IAM & PRIVILEGE RING ISOLATION
Separate IAM least-privilege policies for state tables vs. vector indexes.

DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

Atomic CAS Mutex:
DynamoDB Conditional PutItem (attribute_not_exists) prevents concurrent write collision.

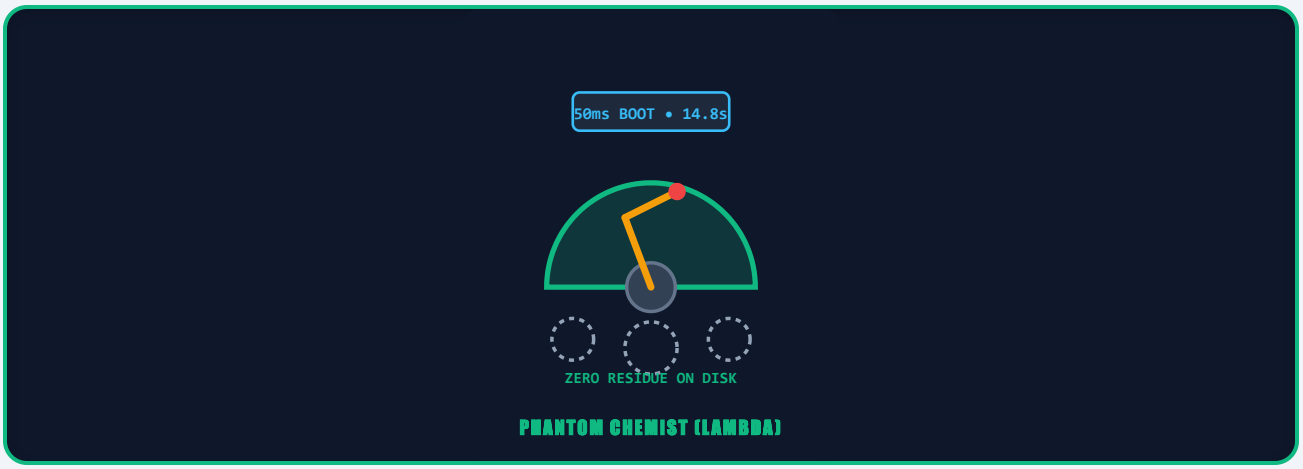
OpenSearch k-NN:
Serverless vector index cosine distance evaluation for certified semantic recall.

Neptune Graph Engine:
Strict relational triple store for entity verification and ontology validation.

⚠ **CATASTROPHIC FAILURE MODE:** Vector search similarity hallucinations treated as authoritative financial state.

HARDWARE CONTAINMENT: Strict architectural partition: state mutations require deterministic DynamoDB CAS locks.

TIER 0 • SKILL #5 — Ring 3



🧪 The Phantom Chemist

AWS Lambda

The Ephemeral 50ms Pop-Up Lab

> *"Treat compute runtimes like paper napkins, not pets. Execute, collect results, and destroy!"*

⚡ **Fresh, pristine compute environments on every call with zero persistent attack surface.**

Formal Engineering Contract & Invariants

SKILL.md Module: `aws-lambda`

⏱ **SLA: Cold start boot < 50ms, execution budget max 14.8s, memory ceiling 512MB.**

SYSTEM ROLE & RESPONSIBILITY

Firecracker MicroVM Ephemeral Executor

INPUT / OUTPUT BOUNDARY CONTRACT

Takes event payload → Boots container in <50ms → Returns stdout result → Destroys runtime.

IAM & PRIVILEGE RING ISOLATION

Scoped execution role with zero persistent local disk write persistence.

DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

Firecracker MicroVM:
Sub-5ms ephemeral container boot with dedicated Linux kernel and jailed cgroups.

Memory / Clock Budget:
Hard wall-clock timeout (max 15.0s) and memory ceiling (512MB) enforced by hypervisor.

Zero Residual State:
Ephemeral disk scrubbed upon container freeze; zero data leakage across invocations.

⚠ **CATASTROPHIC FAILURE MODE:** Memory leaks, poisoned runtime variables, or persistent malware surviving across execution calls.

HARDWARE CONTAINMENT: Hypervisor destroys container filesystem on termination; zero residual state survives.

TIER 0 • SKILL #6 — Ring 0



The Rail Conductor

AWS Step Functions

The Unskippable Ratchet Rail Conductor

> *"You cannot skip to Step 3! The safety ratchet only clicks forward upon valid schema!"*

⚡ **Guarantees workflows never skip prerequisites, drop transactions, or lose execution state.**

Formal Engineering Contract & Invariants

SKILL.md Module: `aws-stepfunctions`

⏱ **SLA: State transition overhead < 15ms in Express Workflows.**

SYSTEM ROLE & RESPONSIBILITY

Deterministic State Machine DAG Orchestrator

INPUT / OUTPUT BOUNDARY CONTRACT

Consumes execution state JSON → Evaluates ASL state transitions → Dispatches task workers.

IAM & PRIVILEGE RING ISOLATION

Task-level IAM role assumption per state transition.

⬇ **DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE** — Root Cloud Primitives & Hardware Proof

Unskippable State DAG:
State transition JSON graph enforced by AWS orchestration engine; zero step skipping.

Heartbeat Watchdogs:
HeartbeatSeconds and TimeoutSeconds fire automated task retry or supervisor kill signal.

Guaranteed Compensation:
Automatic Catch/Retry blocks roll back distributed state to safe compensation checkpoint.

⚠ **CATASTROPHIC FAILURE MODE:** Agents improvising arbitrary workflows and skipping critical safety verifications.

HARDWARE CONTAINMENT: Cloud-managed state machine rejects unverified transitions; impossible to bypass DAG barriers.

TIER 0 • SKILL #7

Ring 0



The Silent Courier

Amazon EventBridge

The Targeted Airport Pager

"Whisper to specific subscribers; never shout raw messages across a swarm!"

⚡ **High-throughput asynchronous communication with zero broadcast chatter or message storms.**

Formal Engineering Contract & Invariants

SKILL.md Module: `amazon-eventbridge`

⏱ **SLA: Event routing latency < 25ms under 10,000 events/sec load.**

SYSTEM ROLE & RESPONSIBILITY

Content-Based Event Bus Router

INPUT / OUTPUT BOUNDARY CONTRACT

Takes cloud event JSON envelope → Matches declarative pattern rules → Routes to target queue.

IAM & PRIVILEGE RING ISOLATION

Target queues require explicit resource policy granting EventBridge publish permissions.

DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

Pattern Filtering:
Declarative content-based rules filter on detail-type and JSON body fields.

Event Archive:
Immutable EventBridge Archive preserves full replayable event stream for forensics.

Dead Letter Queue:
Poison pill or malformed payloads routed to quarantine SQS DLQ; swarm protected.

⚠ **CATASTROPHIC FAILURE MODE:** Broadcast noise storms (50 agents pinging each other causing system buffer collapse).

HARDWARE CONTAINMENT: Hardware-level declarative pattern matching routes messages exclusively to subscribed endpoints.

TIER 1 • SKILL #8 Ring 0



⚖ Judge Prefrontal

`Kernel Propose-Decide`

The Prefrontal Cortex Firewall

> *"Action = Verdict(Proposal). An LLM proposes ideas; the Kernel issues physical verdicts!"*

⚡ **Eliminates rogue AI hallucinations by mathematically separating proposal generation from execution authority.**

Formal Engineering Contract & Invariants

⏱ **SLA: Verdict compilation < 0.8ms.**

SKILL.md Module: `kernel-propose-decide`

SYSTEM ROLE & RESPONSIBILITY
Executive Proposal & Verdict Arbiter

INPUT / OUTPUT BOUNDARY CONTRACT
Takes Ring 3 Model Proposal → Evaluates Ring 0 Cedar Rules → Issues Signed Execution Token.

IAM & PRIVILEGE RING ISOLATION
Only Judge Prefrontal holds the private key to sign execution envelopes.

↓ DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE Root Cloud Primitives & Hardware Proof

Tier 1 Kernel Control:
Ring 0 Prefrontal Judge strips write credentials from LLM; only validated proposals pass.

Tier 0 AWS Primitive:
AWS IAM Session Policies enforce zero 'Allow' on mutation APIs for Bedrock worker role.

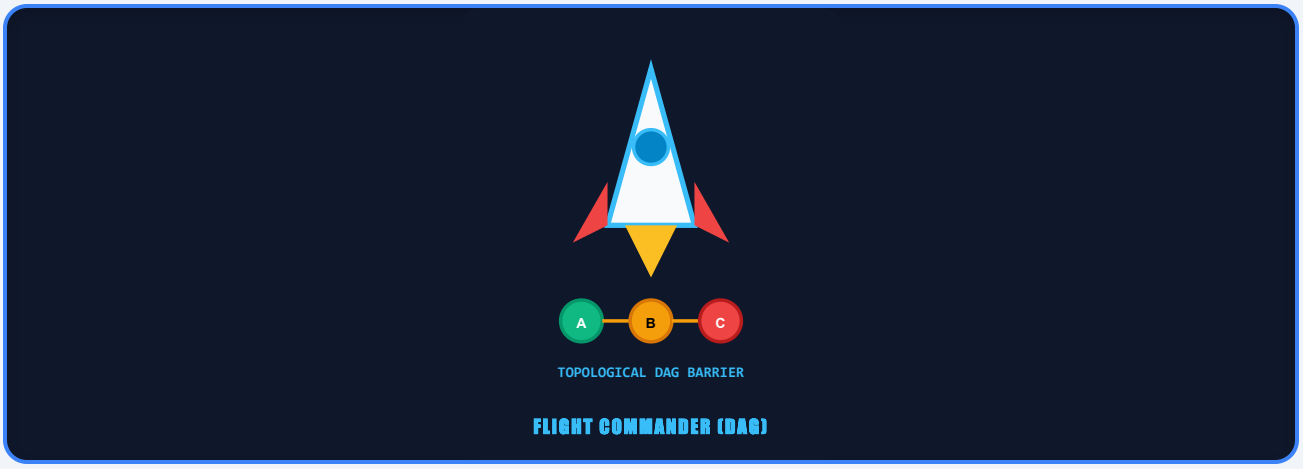
Tier 0 Storage Proof:

Verified proposal token pushed to Amazon EventBridge event bus with KMS HMAC signature.

⚠ **CATASTROPHIC FAILURE MODE:** Rogue hallucination attempting unvetted database mutations (e.g. DROP TABLE prod;).

HARDWARE CONTAINMENT: Complete physical lack of execution credentials. Proposals must pass through Ring 0 Judge Prefrontal.

TIER 1 • SKILL #9 Ring 0



🚀 Flight Commander Checklist

`Kernel Runbook Engine`

The Mission Flight Checklist Commander

"NASA astronauts don't launch rockets by improvising! Step barriers are locked!"

⚡ **Transforms chaotic multi-agent goals into deterministic, reproducible engineering pipelines.**

Formal Engineering Contract & Invariants

SKILL.md Module: `kernel-runbook-engine`

⏱ **SLA: Step barrier transition validation < 1.2ms.**

SYSTEM ROLE & RESPONSIBILITY

Topological Runbook DAG Orchestrator

INPUT / OUTPUT BOUNDARY CONTRACT

Takes declarative runbook YAML → Validates topological DAG → Enforces step barrier gates.

IAM & PRIVILEGE RING ISOLATION

Each DAG node executes with dedicated scoped IAM session credentials.

⬇ **DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE** Root Cloud Primitives & Hardware Proof

Tier 1 Kernel Control:	Tier 0 AWS Primitive:

Flight Checklist parser validates task schema, prerequisites, and idempotent rollback plan.

AWS Step Functions executes immutable ASL state machine; unskippable step order.

Tier 0 MicroVM Node:
AWS Lambda runs isolated atomic tasks in Firecracker microVMs with strict memory caps.

⚠ **CATASTROPHIC FAILURE MODE:** Agents improvising arbitrary workflows and skipping critical safety verifications.

HARDWARE CONTAINMENT: DAG barriers enforced by orchestrator; impossible to proceed to Node B without proof of Node A completion.

TIER 1 • SKILL #10

Ring 0



Sparky the Bulldog

Kernel Supervisor

The Deadlock Watchdog & Kill Switch

"Deadlock detected! Time is up, buster! Container terminated before you burned a million tokens!"

⚡ **Prevents runaway LLM loops and runaway token burns before they consume system resources.**

Formal Engineering Contract & Invariants

SKILL.md Module: `kernel-supervisor`

⏱ **SLA: Hard 15.0s wall-clock countdown; interrupt triggered in < 5ms.**

SYSTEM ROLE & RESPONSIBILITY
Execution Watchdog & Out-of-Band Breaker

INPUT / OUTPUT BOUNDARY CONTRACT
Monitors active thread execution time & memory → Fires SIGKILL if budget ceiling exceeded.

IAM & PRIVILEGE RING ISOLATION
Holds root host container administration privileges.

⬇️ **DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE** Root Cloud Primitives & Hardware Proof

Tier 1 Kernel Control:
Supervisor thread monitors process heartbeat; kills rogue loops or memory leaks.

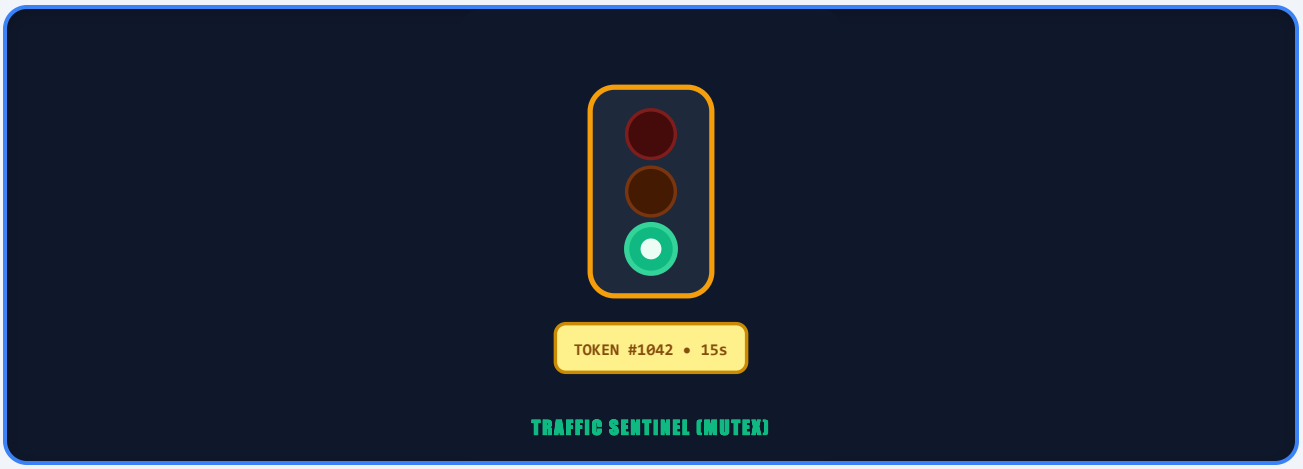
Tier 0 AWS Primitive:
AWS Lambda hypervisor enforces hard 15.0s TimeoutSeconds timer and memory cgroups.

Tier 0 Panic Route:
Amazon CloudWatch Alarm trips on timeout; dispatches SIGKILL and dumps state to S3 DLQ.

⚠️ **CATASTROPHIC FAILURE MODE:** LLM spinning in infinite circular clarification loops, burning millions of tokens and locking threads.

🛡️ **HARDWARE CONTAINMENT:** External timer operating on separate clock circuit terminates container forcefully if lease expires.

TIER 1 • SKILL #11 Ring 0



🚦 The Traffic Sentinel

Kernel Mutex Lease

The Traffic Light & Monotonic Fence Master

"Car B attempts write with old token #1041: REJECTED! Stale out-of-order writes vaporized!"

⚡ **Eliminates race conditions, write collisions, and deadlocks across concurrent agent clusters.**

Formal Engineering Contract & Invariants

⏱️ **SLA: CAS acquisition latency < 3.5ms.**

SKILL.md Module: `kernel-mutex-lease`

SYSTEM ROLE & RESPONSIBILITY
Distributed CAS Lock & Lease Coordinator

INPUT / OUTPUT BOUNDARY CONTRACT
Takes write lock request → Evaluates DynamoDB CAS →

IAM & PRIVILEGE RING ISOLATION
Partition-level write lease keys with cryptographic signature validation.

Emits Monotonic Fence Token with 15s TTL.

DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

Tier 1 Kernel Control:
`Ring 0 Mutex coordinator grants temporary lease token with strictly fenced lease ID.`

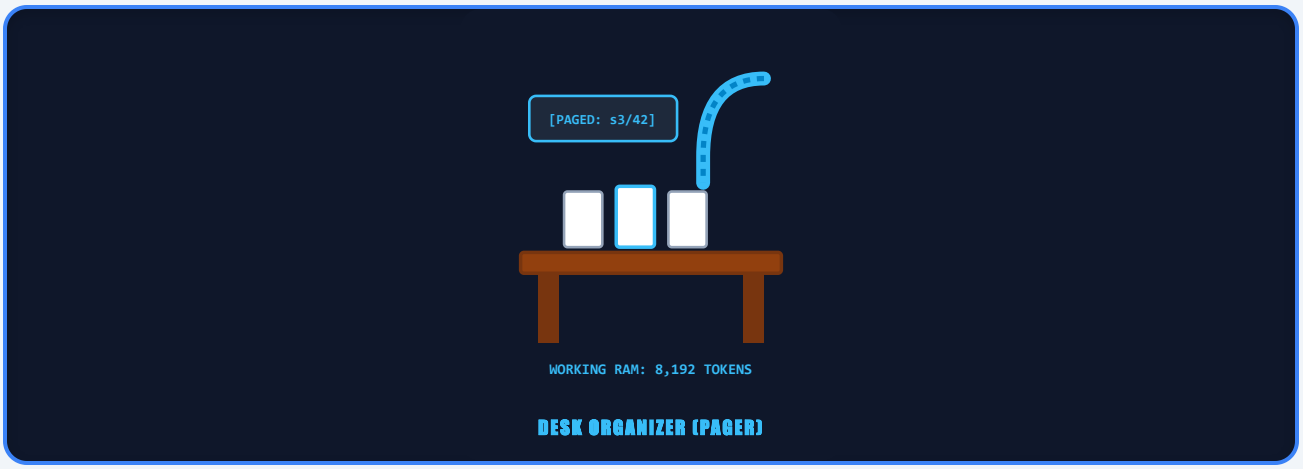
Tier 0 AWS Primitive:
`Amazon DynamoDB PutItem executes atomic CAS: attribute_not_exists(lock) OR expiry < :now.`

Tier 0 TTL Reaping:
`DynamoDB Time-To-Live hardware thread auto-releases lock if agent node goes silent.`

⚠ **CATASTROPHIC FAILURE MODE:** Two concurrent agents trying to write to the same account row simultaneously, causing race condition data corruption.

HARDWARE CONTAINMENT: CAS atomicity guarantees only one winner; stale tokens automatically rejected at database storage boundary.

TIER 1 • SKILL #12 — Ring 0



Desk Organizer

`Kernel Context Pager`

The Virtual RAM Desk Organizer

> *"Never stuff 400 rambling pages into an active prompt! Keep working RAM tiny (8k) and page out history!"*

⚡ **Maintains sharp, hallucination-free reasoning by keeping working context small while preserving infinite memory.**

Formal Engineering Contract & Invariants

SKILL.md Module: `kernel-context-pager`

⏱ **SLA: Page out < 18ms; page in retrieval < 22ms.**

SYSTEM ROLE & RESPONSIBILITY

Virtual Attention RAM & Paging Engine

INPUT / OUTPUT BOUNDARY CONTRACT

Monitors prompt token budget (8,192 max) → Pages cold frames to S3 stubs in <25ms.

IAM & PRIVILEGE RING ISOLATION

Encrypted cache partitions partitioned by agent conversation session ID.

DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE Root Cloud Primitives & Hardware Proof

Tier 1 Kernel Control:
Context Pager prunes transient tokens; pages active working memory into structured KV stores.

Tier 0 AWS Primitive:
Amazon ElastiCache Redis / DynamoDB DAX stores active turn vector cache in <1ms.

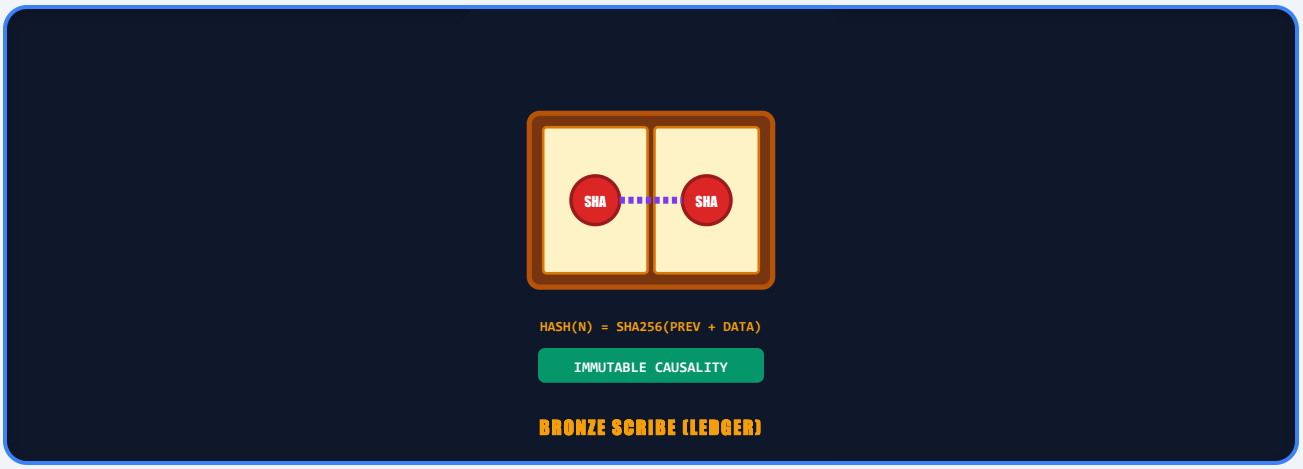
Tier 0 Cold Vault:
Amazon S3 Standard / Glacier Instant Retrieval stores full episodic dialogue history.

⚠ **CATASTROPHIC FAILURE MODE:** Context dilution and needle-in-a-haystack degradation caused by prompt bloat (massive transcripts diluting reasoning).

HARDWARE CONTAINMENT: Automatic context pruner flushes inactive state to persistent storage, preserving sharp reasoning.

TIER 1 • SKILL #13

Ring 0



📜 The Bronze Scribe

Kernel Episodic Ledger

The Blockchain of Agent Cognition

"Block(N) = SHA256(Block(N-1) + Data). Altering one comma breaks the entire chain!"

⚡ **Mathematically guarantees that agent decision history can never be rewritten or falsified.**

Formal Engineering Contract & Invariants

SKILL.md Module: `kernel-episodic-ledger`

⏱ SLA: Cryptographic block chaining overhead < 0.6ms.

SYSTEM ROLE & RESPONSIBILITY

Cryptographic Episodic Audit Chainer

INPUT / OUTPUT BOUNDARY CONTRACT

Takes perception vector + action decision → Computes SHA-256 block diff → Appends to Merkle chain.

IAM & PRIVILEGE RING ISOLATION

Append-only storage permission; overwrite mathematically rejected.

⬇️ DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

Tier 1 Kernel Control:
`Epistemic Ledger links block diffs: Hash(N) = SHA256(Block_N + Hash_N-1) per agent.`

Tier 0 AWS Primitive:
`Amazon S3 Object Lock (Compliance Mode) locks block digest; impossible to overwrite.`

Tier 0 CloudTrail Key:
`AWS KMS HMAC-SHA256 signs ledger root; CloudTrail streams event to immutable SIEM.`

⚠️ **CATASTROPHIC FAILURE MODE:** Rogue agent or malicious actor attempting to quietly delete or edit past mistakes from audit logs.

🛡️ **HARDWARE CONTAINMENT:** Mathematical properties of SHA-256 hash chains make any retroactive alteration immediately detectable.

TIER 1 • SKILL #14

Ring 0



🔍 Detective Scoop

Kernel Grounding Certifier

The 3-Source Investigative Reporter

"Hold your horses, kid! We don't publish rumors—all THREE independent proofs must agree before I stamp this paper!"

⚡ **Completely annihilates hallucinated assertions before they enter the swarm decision pipeline.**

Formal Engineering Contract & Invariants

SKILL.md Module: `kernel-grounding-certifier`

⏱ **SLA: Tri-store verification parallel query < 16ms.**

SYSTEM ROLE & RESPONSIBILITY
Tri-Store Fact Triangulation Engine

INPUT / OUTPUT BOUNDARY CONTRACT
Takes candidate proposition → Queries S3 receipt, vector cosine (>=0.82), and Neptune graph triple → Stamps CERT_OK.

IAM & PRIVILEGE RING ISOLATION
Read-only access across certified corporate evidentiary stores.

⬇️ **DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE** — Root Cloud Primitives & Hardware Proof

Tier 1 Kernel Control:
Investigative Reporter demands 3 independent corroborating proofs before action certification.

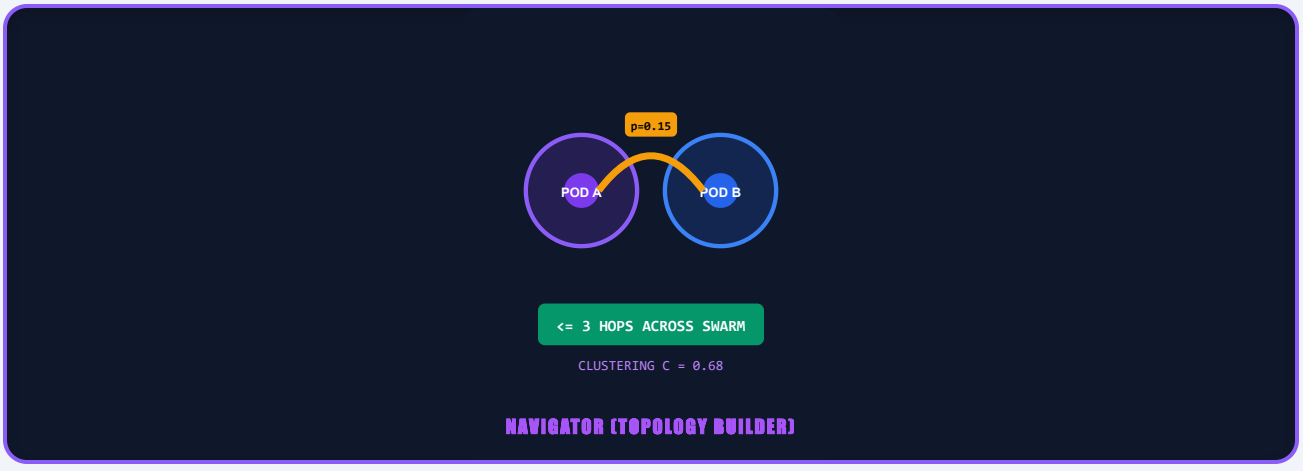
Tier 0 AWS Primitive:
Amazon OpenSearch / Bedrock Knowledge Bases performs hybrid vector + BM25 citation match.

Tier 0 Attestation:
AWS Nitro Enclave cryptographically verifies citation cosine similarity before sign-off.

⚠️ **CATASTROPHIC FAILURE MODE:** Unchecked LLM hallucinations entering the enterprise decision stream as factual truth.

🛡️ **HARDWARE CONTAINMENT:** Tri-store consensus protocol annihilates ungrounded assertions before they reach the execution pipeline.

TIER 2 • SKILL #15 — Ring 0



🌐 Air Traffic Navigator

Swarm Topology Builder

The Small-World Airport Hub Network

"Clustering coefficient C=0.68! Any agent reaches any other in <= 3 hops without broadcast storms!"

⚡ **Connects large swarms of agents with ultra-short communication paths while eliminating O(N^2) message storms.**

Formal Engineering Contract & Invariants

SKILL.md Module: `swarm-topology-builder`

⏱ **SLA: Path resolution < 0.4ms across 100-node graph.**

SYSTEM ROLE & RESPONSIBILITY
Watts-Strogatz Small-World Topology Orchestrator

INPUT / OUTPUT BOUNDARY CONTRACT
Takes N agent definitions → Computes k-regular ring lattice -> Injects p=0.15 shortcut bridges.

IAM & PRIVILEGE RING ISOLATION
Inter-pod routing policies certified by Ring 0 Cedar ACLs.

⬇ DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

Tier 2 Swarm Lattice:
Constructs small-world mesh with clustering coefficient C ⩾ 0.6 and average path length L ⩽ 3.

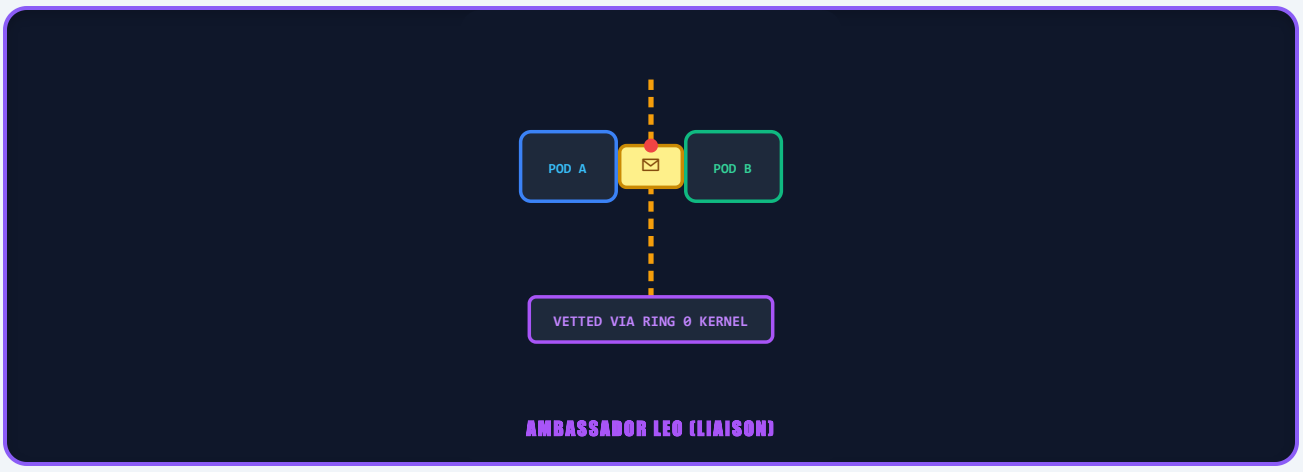
Tier 1 Kernel Gate:
Ring 0 Cedar Policy Gate evaluates inter-agent edge graph: permit(Edge::A, Edge::B).

Tier 0 AWS Primitive:
Amazon DynamoDB Global Tables stores graph adjacency matrix; AWS PrivateLink routes traffic.

⚠ **CATASTROPHIC FAILURE MODE:** Exponential message storms (2,500 msgs/sec) where 50 agents spam each other and choke network buffers.

HARDWARE CONTAINMENT: Mathematically structured topology routes messages through cluster hubs and long-range shortcuts.

TIER 2 • SKILL #16 — Ring 0



Ambassador Leo

Swarm Liaison Router

The Diplomatic Embassy Channel

> *"Internal chatter stays inside specialist pods! Cross-pod queries travel through designated ambassadors and pass through Ring 0!"*

⚡ **Maintains strict modularity across agent domains, preventing communication leaks and cross-pod confusion.**

Formal Engineering Contract & Invariants

⏱ **SLA: Cross-pod bridge latency < 4.2ms.**

SKILL.md Module: `swarm-liaison-router`

SYSTEM ROLE & RESPONSIBILITY

Cross-Pod Embassy Liaison Router

INPUT / OUTPUT BOUNDARY CONTRACT

Intercepts cross-pod messages → Packages into formal diplomatic envelope → Vets through Ring 0.

IAM & PRIVILEGE RING ISOLATION

Cross-boundary VPC endpoint security group authorization.

⬇ DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

Tier 2 Swarm Lattice:

Liaison Agent bridges Demand Pod, Supply Pod, and Control Pod via dedicated embassy channels.

Tier 1 Kernel Gate:

Ring 0 Cedar evaluates cross-pod boundary rule; Ring 3 Epistemic Ledger logs the border crossing.

Tier 0 AWS Primitive:

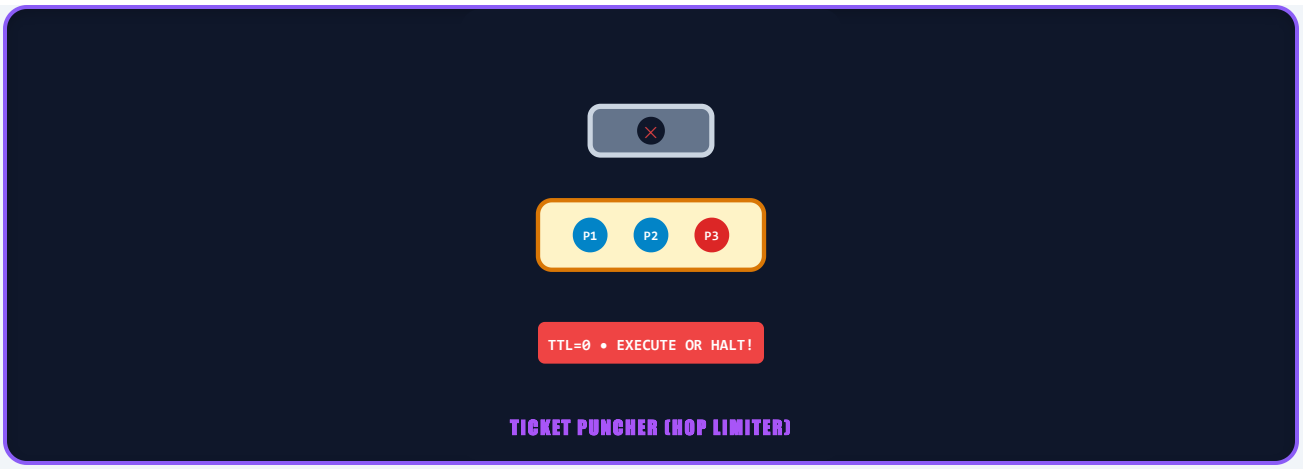
AWS VPC Peering / Transit Gateway routes packets across VPCs; AWS KMS re-encrypts envelopes.

⚠ **CATASTROPHIC FAILURE MODE:** Cross-domain contamination (e.g. 67M citizens texting each other directly across national borders).

HARDWARE CONTAINMENT: Pod isolation enforced by network security groups; cross-pod traffic restricted to accredited ambassador interfaces.

TIER 2 • SKILL #17

Ring 0



The Ticket Puncher

Swarm Hop Limiter

The 3-Hop Game of Hot Potato Conductor

> *"OUT OF TICKETS! Max 3 hops reached! No passing the buck—execute the task NOW or Sparky cuts the power!"*

⚡ **Permanently terminates infinite delegation loops and circular gossip tag games in multi-agent networks.**

Formal Engineering Contract & Invariants

⏱ **SLA: Ticket punch inspection < 0.1ms.**

SKILL.md Module: `swarm-hop-limiter`

SYSTEM ROLE & RESPONSIBILITY

Bounded Hop TTL & Circular Delegation Guard

INPUT / OUTPUT BOUNDARY CONTRACT

Attaches TTL=3 header to delegation packet → Decrements on forwarding → Terminates thread if TTL=0.

IAM & PRIVILEGE RING ISOLATION

Immutable visited-node bitmap header.

DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

Tier 2 Swarm Lattice:

Attaches Hop Count = 3 metadata to all peer-to-peer inter-agent messages across the swarm.

Tier 1 Kernel Gate:

Ring 1 Supervisor decrements TTL; if Hop Count reaches 0, Supervisor halts thread before LLM call.

Tier 0 AWS Primitive:

Amazon SQS Dead Letter Queue (DLQ) captures expired packet; Step Functions limits DAG recursion.

⚠ **CATASTROPHIC FAILURE MODE:** Lazy agents passing tasks in infinite circular delegation games ('Not my job, forwarding!').

HARDWARE CONTAINMENT: Monotonic decrement counter trips hard interrupt when TTL=0, forcing execution or quarantine.

TIER 2 • SKILL #18

Ring 0



📋 Tactical Quartermaster

Swarm Shared Whiteboard

The Tactical Combat Blackboard Quartermaster

> *"Enemy contact bearing 270! I write it once to the Operations Lane; all 30 officers see it simultaneously in real time!"*

⚡ **Enables dozens of agents to collaborate with O(N) write complexity instead of O(N^2) communication explosion.**

Formal Engineering Contract & Invariants

⏱ **SLA: Lane broadcast update < 3.8ms.**

SKILL.md Module: `swarm-shared-whiteboard`

SYSTEM ROLE & RESPONSIBILITY

Partitioned O(N) Whiteboard Coordinator

INPUT / OUTPUT BOUNDARY CONTRACT

Maintains partitioned blackboard lanes (Demand, Supply, Control) with vector clocks.

IAM & PRIVILEGE RING ISOLATION

Lane-level write ownership enforced by CAS Mutex Talking Stick.

⬇️ DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE

Root Cloud Primitives & Hardware Proof

Tier 2 Swarm Lattice:

Maintains shared blackboard with partitioned lanes (Demand, Supply, Control) and vector clocks.

Tier 1 Kernel Gate:

Ring 2 Context Pager loads active lane slices; Ring 0 Cedar enforces lane write permissions.

Tier 0 AWS Primitive:

Amazon DynamoDB PutItem executes row-level CAS locks; Amazon Kinesis replicates updates across pods.

⚠ **CATASTROPHIC FAILURE MODE:** Communication explosion where N agents require N(N-1)/2 messages to synchronize state.

HARDWARE CONTAINMENT: Centralized partitioned memory guarantees linear O(N) communication complexity.

TIER 2 • SKILL #19

Ring 0



Chief Flight Quorum Director

Swarm Consensus Arbiter

The Byzantine Spaceflight Quorum Director

> *"Supermajority >= 10 of 12 votes reached! Faulty ballots discarded! Trajectory maintained flawlessly!"*

⚡ **Reaches deterministic, tamper-proof consensus even when up to 3 agent nodes glitch, crash, or hallucinate.**

Formal Engineering Contract & Invariants

SKILL.md Module: `swarm-consensus-arbiter`

⏱ **SLA: Byzantine consensus voting round < 4.5ms.**

SYSTEM ROLE & RESPONSIBILITY

Byzantine Fault-Tolerant Consensus Arbiter

INPUT / OUTPUT BOUNDARY CONTRACT

Collects N signed cryptographic ballots → Tally quorum (>= 2f+1) → Emits certified majority resolution.

IAM & PRIVILEGE RING ISOLATION

Digital ballot signatures verified against multi-region KMS public keys.

⬇ **DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE** — Root Cloud Primitives & Hardware Proof

Tier 2 Swarm Lattice:
Collects independent cryptographic ballots across peer agents; demands supermajority consensus.

Tier 1 Kernel Gate:
Ring 0 Cedar verifies 2f+1 signed ballot tokens; Ring 3 Epistemic Ledger signs combined block.

Tier 0 AWS Primitive:
AWS Nitro Enclaves executes multi-party ballot tally; AWS KMS Multi-Region keys verify signatures.

⚠ **CATASTROPHIC FAILURE MODE:** A glitched or compromised node screaming erratic commands ('Fire all thrusters into the sun!').

HARDWARE CONTAINMENT: Supermajority mathematical quorum outvotes rogue nodes in sub-5ms intervals.

TIER 2 • SKILL #20

Ring 0



Marshal Redcard

Swarm Circuit Breaker

The Formula 1 Red Card & Pit Stop Marshal

"3 strikes reached! Snatched your talking stick! 5-minute cooldown in the pit lane while Lambda deploys a fresh replacement!"

⚡ **Prevents a single malfunctioning agent from creating a domino effect that crashes the entire swarm.**

Formal Engineering Contract & Invariants

SKILL.md Module: `swarm-circuit-breaker`

⏱ **SLA: Circuit breaker trip & mutex revocation < 8ms.**

SYSTEM ROLE & RESPONSIBILITY
Swarm Cascade Quarantine & Mutex Revocation Sentinel

INPUT / OUTPUT BOUNDARY CONTRACT
Tracks 3-strike anomaly pattern → Forcefully strips mutex lease

IAM & PRIVILEGE RING ISOLATION
Emergency administration role with credential detachment authority.

→ Benches agent node to pit lane.

DOWNWARD TRACEABILITY TO TIER 0 AWS INFRASTRUCTURE — Root Cloud Primitives & Hardware Proof

Tier 2 Swarm Lattice:
Detects erratic agent behavior (3 consecutive errors/duplicate POs); raises emergency red card.

Tier 1 Kernel Gate:
Ring 1 Supervisor forcefully strips active Mutex lease token; sends SIGKILL to container cgroup.

Tier 0 AWS Primitive:
Amazon EventBridge fires automated IAM detachment lambda; DynamoDB forcefully resets row mutex.

⚠ **CATASTROPHIC FAILURE MODE:** A single malfunctioning agent causing a cascading domino crash across the entire 50-agent swarm.

HARDWARE CONTAINMENT: Rapid circuit breaker trips in <50ms, revoking credentials and quarantining failing pods.